Does the value of living space vary by dwelling type?

[Team member names]

STA302 Part 1 - due November 5, 2026

Working draft: confirm Ames eligibility; replace team and Module 3 placeholders.

Contributions

[Member A]: [actual contribution]. [Member B]: [actual contribution]. [Member C]: [actual contribution]. [Member D]: [actual contribution].

Introduction

Among residential properties sold in Ames, Iowa, during 2006-2010, does the association between above-ground living area and sale price differ by dwelling type, after adjustment for lot area, construction quality, age, basement area, garage capacity, neighborhood, and sale year? Sale price is the recorded transaction amount in nominal US dollars. Rather than assuming a universal price-per-square-foot relationship, we estimate the adjusted sale-price difference associated with 100 additional square feet within each dwelling type. These estimates describe associations, not the causal return from enlarging a home.

Housing attributes need to be considered jointly, and past research highlights several key variables. Mathur (2019), studying King County, Washington, found that construction quality and maintenance were associated with higher prices. A medium-quality home sold for approximately 25% more than a low-quality home. This supports adjustment for quality, while the maintenance findings motivate considering condition in subsequent models. Goodman and Thibodeau (1995) analyzed nearly 8,500 Dallas single-family transactions. They found nonlinear depreciation and age-related heteroskedasticity. These findings motivate including age and checking both functional form and residual variance. Fik et al. (2003), studying the Tucson housing market, found that location and housing attributes jointly influenced prices in ways that simple location indicators did not fully capture. This motivates neighborhood adjustment, while recognizing that spatial dependence may remain.

Together, these studies support an adjusted analysis while warning against a universal price-per-square-foot rule. Our extension asks whether the living-area slope varies across five recorded dwelling types in Ames. We hypothesize that differences in layout and shared structures make a common slope too restrictive. We propose this as a local extension to Ames, not as a claim that dwelling-type interactions have never been studied.

Multiple linear regression estimates conditional mean-price differences while adjusting for several characteristics simultaneously. A correlation or comparison of group means cannot isolate the adjusted area association or test whether slopes differ. Type-specific slopes, confidence intervals, and a joint partial F-test of the four interaction coefficients answer the question. The outcome is sufficiently continuous for this approach, subject to diagnostic assessment. Historical housing researchers and appraisers studying comparable-property adjustments could benefit. However, these estimates are not current valuations and should not be transferred directly to other markets or time periods.

Data description

We downloaded the original tab-delimited data and codebook from the Journal of Statistics Education archive (De Cock, 2011). The Ames City Assessor's Office originally recorded property characteristics and sales for valuation and assessment. De Cock obtained a records-system extract, removed specialized assessment variables and nonresidential

records, and retained the latest transaction for repeatedly sold properties. His public release contains 2,930 properties and 82 fields and was curated for regression teaching.

The analysis uses 2,928 complete records, excluding one missing basement area and one missing garage capacity. Sale price has mean \$180,841, median \$160,000, standard deviation \$79,890, and range \$12,789-\$755,000. Its many possible values support a continuous response; right skew does not itself invalidate regression, whose assumptions concern conditional errors. Unique parcel identifiers reduce repeated-property dependence, but neighboring homes and common market conditions can still produce dependent errors.

Nine predictors enter the model: living area, dwelling type, lot area, overall quality, age at sale, basement area, garage capacity, neighborhood, and sale year. Dwelling type and neighborhood are nominal; sale year is a factor. Quality is provisionally treated as a numerical 1-10 score, imposing equal increments. Living area is measured per 100 square feet and centered at 1,500 square feet; lot area is per 1,000 square feet and basement area per 100 square feet. These unit changes do not repair model assumptions.

The living-area-by-dwelling-type interaction allows different slopes for detached homes, converted two-family homes, duplexes, and interior/end-unit townhouses. These are existing categories, so no arbitrary numerical cutoffs are needed. Figure 2 displays the adjusted slopes; differing layouts provide the substantive motivation.

Table 1. Complete-case summaries. Prices and areas have long upper tails. Original missing counts: basement area 1, garage capacity 1; all other selected variables 0. One age equals -1; retain for investigation. Dwelling types are imbalanced; Landmrk has one observation.

Variable	N	Mean	SD	Min	Q1	Median	Q3	Max
SalePrice	2928	180841.03	79889.90	12789.00	129500.00	160000.00	213500.00	755000.00
Living100	2928	15.00	5.06	3.34	11.26	14.42	17.42	56.42
Lot1000	2928	10.15	7.88	1.30	7.44	9.44	11.56	215.24
Quality	2928	6.10	1.41	1.00	5.00	6.00	7.00	10.00
Age	2928	36.41	30.29	-1.00	7.00	34.00	54.00	136.00
Basement100	2928	10.52	4.41	0.00	7.93	9.90	13.02	61.10
GarageCars	2928	1.77	0.76	0.00	1.00	2.00	2.00	5.00

Variable	Level	N	Percent
Building	1Fam	2423	82.75
Building	2fmCon	62	2.12
Building	Duplex	109	3.72
Building	Twnhs	101	3.45
Building	TwnhsE	233	7.96
Neighborhood	NAmes	443	15.13
Neighborhood	Blmngtn	28	0.96
Neighborhood	Blueste	10	0.34
Neighborhood	BrDale	30	1.02
Neighborhood	BrkSide	107	3.65
Neighborhood	ClearCr	44	1.50
Neighborhood	CollgCr	267	9.12
Neighborhood	Crawfor	103	3.52
Neighborhood	Edwards	194	6.63
Neighborhood	Gilbert	165	5.64
Neighborhood	Greens	8	0.27
Neighborhood	GrnHill	2	0.07
Neighborhood	IDOTRR	92	3.14
Neighborhood	Landmrk	1	0.03
Neighborhood	MeadowV	37	1.26
Neighborhood	Mitchel	114	3.89
Neighborhood	NoRidge	71	2.42
Neighborhood	NPkVill	23	0.79

Variable	Level	N	Percent
Neighborhood	NridgHt	166	5.67
Neighborhood	NWAmes	131	4.47
Neighborhood	OldTown	239	8.16
Neighborhood	Sawyer	151	5.16
Neighborhood	SawyerW	125	4.27
Neighborhood	Somerst	182	6.22
Neighborhood	StoneBr	51	1.74
Neighborhood	SWISU	48	1.64
Neighborhood	Timber	72	2.46
Neighborhood	Veenker	24	0.82
SaleYear	2006	625	21.35
SaleYear	2007	693	23.67
SaleYear	2008	621	21.21
SaleYear	2009	648	22.13
SaleYear	2010	341	11.65

Variable	Missing
SalePrice	0
Living100	0
Building	0
Lot1000	0
Quality	0
Age	0
Basement100	1
GarageCars	1
Neighborhood	0
SaleYear	0

Ethics discussion

The Ames Housing dataset uses real sales records from the Ames City Assessor's Office. It is not simulated data. De Cock (2011) explains where the data came from, and a codebook is provided to explain the variables. This makes the dataset more trustworthy, but it does not mean the data are perfect. For example, one house has a calculated age of -1, and the overall quality score may depend on the assessor's judgment. Also, the data only include houses sold in Ames from 2006 to 2010, so the results may not apply to other places.

We also need to consider privacy and consent. The records were originally collected for property assessment, and we cannot confirm whether homeowners agreed to their use in research. Even without names, some property details might be matched with other records. We will not try to identify any owners.

For ownership, the records came from the Assessor's Office and were later published by De Cock. We will acknowledge both and use the dataset only for this project. We will also avoid making unfair judgments about neighborhoods based on differences in house prices.

Preliminary results

The preliminary model (Table 2) regresses sale price on nine predictors plus a living area × dwelling type interaction (adjusted $R^2 = 0.839$; residual SE = \$32,073).

For detached homes, 100 additional square feet is associated with a \$4,860 higher mean price (95% CI \$4,507–\$5,212), holding other predictors fixed. The corresponding slopes are \$2,419 for two-family conversions, \$1,900 for duplexes, \$1,832 for interior townhouses and \$6,042 for end-unit townhouses (Figure 2), and the four interaction terms are jointly significant (partial $F(4, 2882) = 9.20$, $p < 0.001$).

Figure 3 summarizes pairwise associations among the numeric predictors. At high fitted prices, observed prices bend above the identity line (Figure 1a). Residuals curve against fitted price (Figure 1b), living area (1c), and basement area (1g), suggesting that the model misses some nonlinear structure. Constant variance is violated: residual spread increases with fitted price (Figure 1b). Uncorrelated errors appear reasonable: residuals show no trend over sale time (Figure 1l), though nearby homes may share unmeasured influences. Normality is violated: the Q-Q plot (Figure 1m) shows heavy tails. The three largest negative residuals (rows 1499, 2181 and 2182) are partial sales above 4,000 square feet, which De Cock (2011) flags as unlikely to reflect market value; Landmrk has a single sale (leverage one, Figure 1n). No corrections are applied at this stage.

The quality coefficient ($15,672 per point) agrees in direction with Mathur (2019), the negative age coefficient (−$404 per year) is consistent with depreciation reported by Goodman and Thibodeau (1995), though our residual spread is largest for newer homes, and large neighborhood effects echo Fik et al. (2003). Unexpectedly, end-unit townhouses have a larger estimated slope than detached homes, but the difference is not significant (95% CI −$326 to $2,690). GrnHill (2 sales) and Landmrk (1) give unstable estimates.

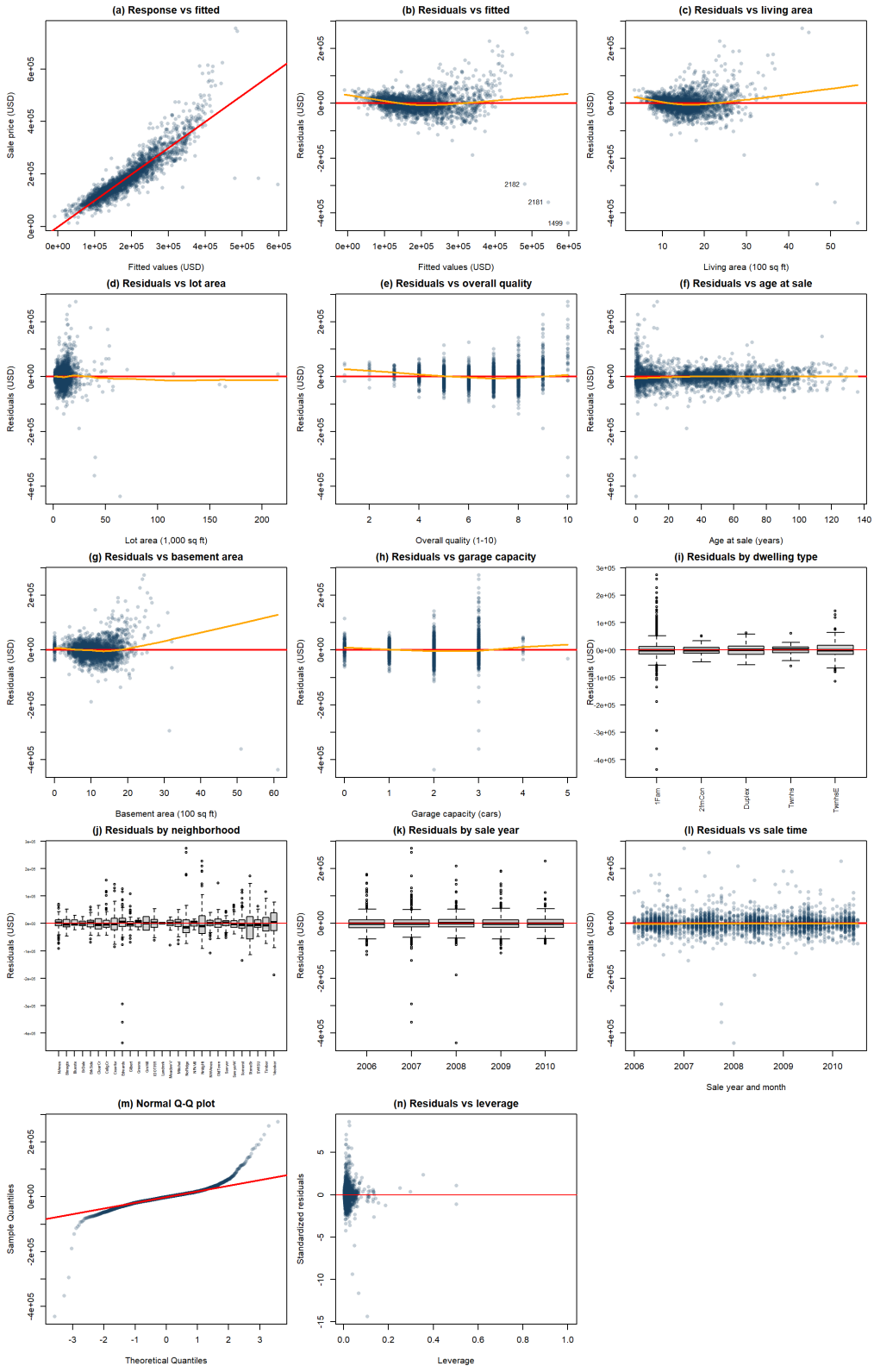


Figure 1: Residual diagnostics for the preliminary sale-price model. Residuals curve and spread out at higher fitted prices (b), and the normal Q-Q plot shows heavy tails (m). Orange curves show local trends. Labels in (b) identify the three most negative residuals by original data row.

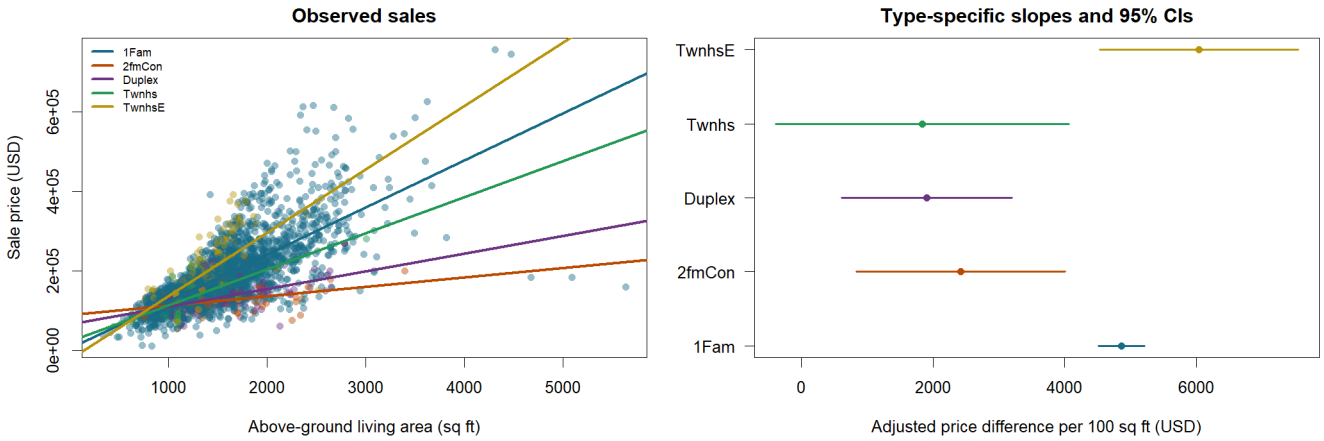


Figure 2: Observed sales with a least-squares line for each dwelling type (left) and adjusted living-area slopes with 95% CIs (right).

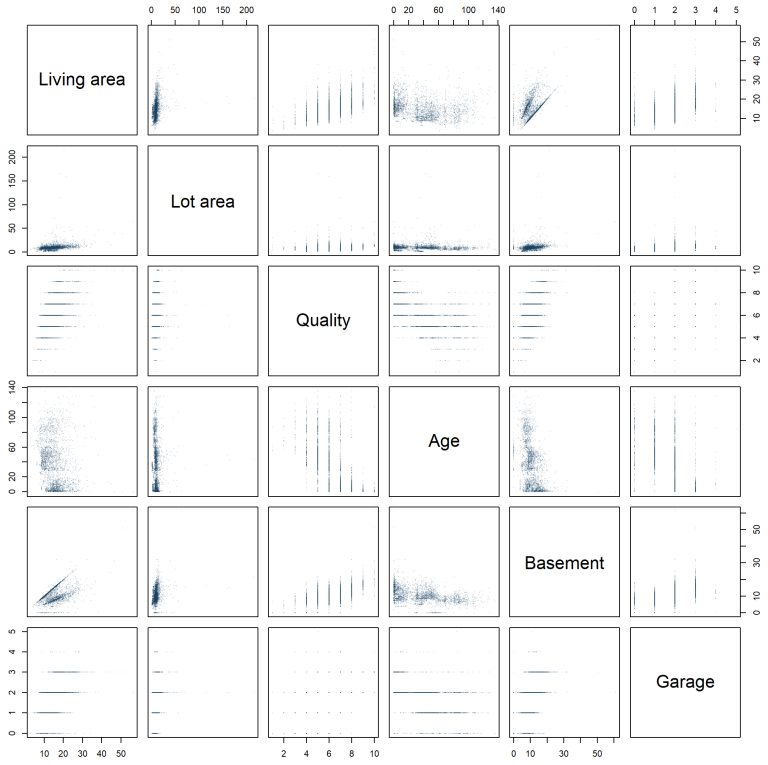


Figure 3: Pairwise scatterplots of the numerical predictors.

Table 2. Preliminary model, USD. References: detached home, North Ames, 2006. Dwelling-type contrasts are at 1,500 sq ft; interactions are slope differences per 100 sq ft. Neighborhood and sale-year effects are included but not shown.

Term	Estimate	SE	95% CI
Intercept	49,004	5,801	[37,630, 60,378]
Living area (per 100 sq ft, centred at 1,500)	4,860	180	[4,507, 5,212]
Type: two-family conversion	-9,749	4,279	[-18,139, -1,360]

Term	Estimate	SE	95% CI
Type: duplex	-21,566	3,426	[-28,285, -14,848]
Type: townhouse (interior)	-46,392	4,828	[-55,860, -36,924]
Type: townhouse (end unit)	-24,725	3,104	[-30,811, -18,639]
Lot area (per 1,000 sq ft)	529	88	[356, 702]
Overall quality (per point)	15,672	765	[14,172, 17,172]
Age at sale (per year)	-404	46	[-493, -314]
Basement area (per 100 sq ft)	2,260	182	[1,903, 2,617]
Garage capacity (per car)	10,229	1,110	[8,052, 12,406]
Living area x two-family conversion	-2,441	820	[-4,050, -833]
Living area x duplex	-2,959	674	[-4,280, -1,638]
Living area x townhouse (interior)	-3,028	1,147	[-5,277, -778]
Living area x townhouse (end unit)	1,182	769	[-326, 2,690]

Plan

Retain living area, dwelling type, and their interaction because they define the question; retain neighborhood and sale year as contextual controls. Preserve all main effects whenever an interaction is included. Compare scientifically justified nested specifications using partial F-tests once assumptions are reasonably supported, and use adjusted R-squared as supporting rather than decisive evidence. Avoid automatic elimination of individual factor levels or selection solely by p-values. Consider adding overall condition and remodeling information, motivated by maintenance research, while checking redundancy among structural measures.

Recheck residual-versus-fitted, Q-Q, scale-location, leverage, and residual-age/time plots after every substantive revision. The fan-shaped spread suggests considering log sale price; curvature suggests quadratic age or living-area terms. Compare diagnostic improvement and interpretability rather than assuming a log transform fixes everything. Investigate the negative age and unusual large-property transactions against the codebook before any correction or exclusion. Assess rare-neighborhood instability, possible collinearity, and neighborhood dependence; discuss methods beyond course scope with the TA. Document sensitivity analyses rather than removing observations to achieve significance.

Proposed schedule (the final submission is at the end of November; confirm the exact date): October 30-31, [Member A] and [Member B] verify sources and data. November 1-3, [Member C] checks code and [Member D] checks interpretation. November 4, all members review; November 5, [Submitter] submits the proposal. November 9-15, [Members A/C] investigate diagnostics and compare revisions. November 16-20, [Members B/D] finalize interpretation and create the poster. November 21-25, all members rehearse and record the presentation. November 26-28, all members audit reproducibility and submission files, leaving a buffer before the confirmed deadline. Combine roles if the team has fewer than four members.

Bibliography

De Cock, D. (2011). Ames, Iowa: Alternative to the Boston housing data as an end of semester regression project. Journal of Statistics Education, 19(3). https://doi.org/10.1080/10691898.2011.11889627

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Data and codebook: https://jse.amstat.org/v19n3/decock/AmesHousing.txt and https://jse.amstat.org/v19n3/decock/DataDocumentation.txt. Accessed October 29, 2026.